

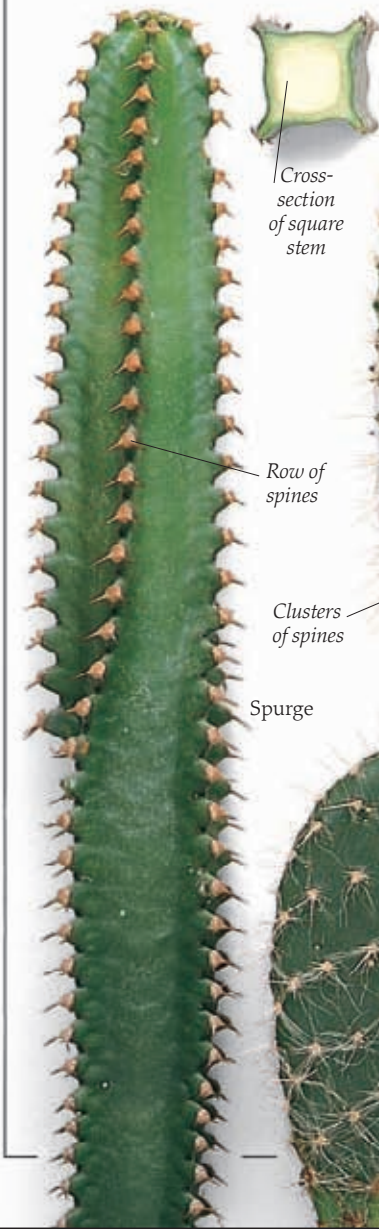
Living without water

NO PLANT CAN LIVE entirely without water. But in very dry regions, where water is scarce, plants called cacti and succulents are able to survive for a number of years between rainstorms. In the world's driest places, rain often comes in irregular but heavy bursts, so the plants that live there have evolved ways of collecting as much water as possible during downpours. The water is then stored in preparation for the next drought. Many cacti and succulents have very long roots. Most of these roots grow near the surface, so that when it rains the plants can collect water from a wide area. Once the water is inside the plant, it is kept there by a number of special adaptations. Plants normally lose water from pores, called stomata, in the surface of their leaves. The plant can control these pores, and it will close them if it begins to lose too much water.

Many cacti and succulents open the pores only at night, when the air is cool and less water can evaporate. Some of these plants have gotten around the problem of water loss by losing their leaves altogether.

SIMILAR SHAPES FOR SIMILAR LIFESTYLES

Not all the spiny plants that live in dry places are cacti. The cactuslike plant on the far left is actually a spurge, a plant quite unrelated to the two cacti to its right. Like the cacti, it has lost its leaves and developed a tough, water-holding stem. This is a typical example of convergent evolution, in which plants or animals in similar environments have evolved in similar ways.

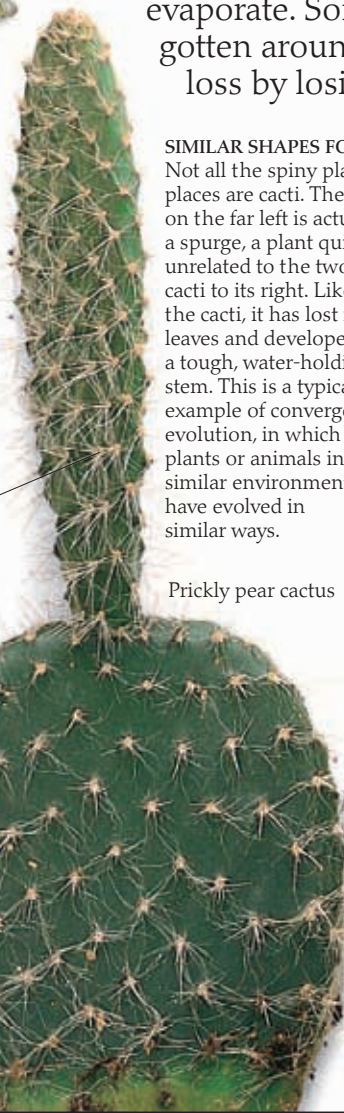


Cross-section of square stem

Row of spines

Clusters of spines

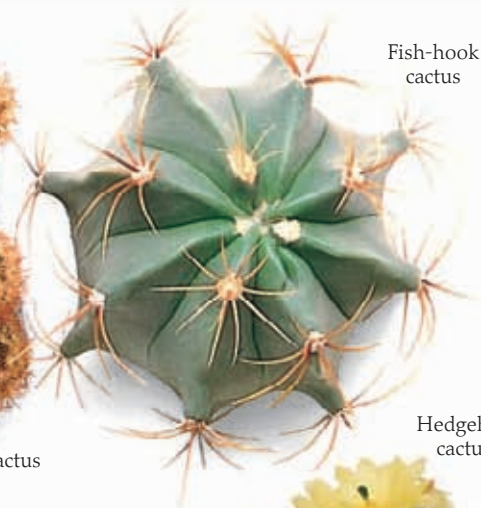
Spurge



Prickly pear cactus



Barrel cactus



Fish-hook cactus

Many cacti have beautiful flowers



Hedgehog cactus

THE CACTUS FAMILY

Almost all true cacti come from the Americas. Because they live in very dry places, they have had to evolve strange shapes to be able to survive. Most cacti have very thick stems and thick groups of spines instead of normal leaves. These spines may protect the plant from heat and cold, as well as from attack by animals. Many cacti have ridges down their stems to allow them to expand and store water when it rains.

Cleistocactus



Cross-section of round stem

THE GIANT SAGUARO

The saguaro is one of the tallest cacti. A 250-year-old plant can be 60 ft (20 m) high and weigh 6½ tons (6 metric tons).

